

Let V be a fin. dim'l v.space.

We saw how to get a finite basis for V using either strategy A or strategy B.

Def: $\left\{ \begin{array}{l} \text{Let } V, W \text{ be vector spaces} \\ \text{A map } V \xrightarrow{f} W \text{ is called an isomorphism if} \end{array} \right.$

o) f is a bijection

1) $f(v_1 + v_2) = f(v_1) + f(v_2)$ " f preserves addition "

2) $f(\lambda v_1) = \lambda f(v_1)$ " ——— scalar mult. "

(for all $v_1, v_2 \in \text{v.space } V$, for all real λ)

Observe • If f is an isomorphism, so is f^{-1}

• If $V \xrightarrow{f} W \xrightarrow{g} U$ are isomorphisms then
so is $V \xrightarrow{g \circ f} U$

Sketch $\left\{ \begin{array}{l} \text{Composition of injective functions is injective.} \\ \text{————— surjective ————— surjective} \end{array} \right.$

$$\left\{ \begin{array}{l} g \circ f(v_1 + v_2) = g(f(v_1 + v_2)) = g(f(v_1) + f(v_2)) = g f(v_1) + g f(v_2) \\ \text{Similarly } g \circ f(\lambda v) = g(\lambda f(v)) = \lambda g f(v) \end{array} \right.$$

→ This part is valid for any $f \in g$ that satisfy ① & ②
(even if they are not bijections).

We say V & W are isomorphic vector spaces (" $V \simeq W$ ")
if there exists an isomorphism $V \xrightleftharpoons[f^{-1}]{f} W$.

Note $V \simeq W \quad W \simeq U \Rightarrow V \simeq U$

Observe If S is a basis for V , then
 $f(S)$ is a basis for W (where $V \xrightarrow{f} W$)

Claim : If $(v_1, \dots, v_n)$ is an ^{ordered} basis for a finite n -space V
then I get an isomorphism $f: \mathbb{R}^n \rightarrow V$

Pf

Define $\mathbb{R}^n \xrightarrow{f} V$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \mapsto \sum_{i=1}^n c_i v_i$$

Such a map can be defined for any list of vectors $v_1, \dots, v_n$ in V .

- Then $\text{Im}(f) = \text{Span}(v_1, \dots, v_n)$
 $\Rightarrow f$ is surjective iff $(v_1, \dots, v_n)$ span V

- f is 1-1 $\Leftrightarrow (v_1, \dots, v_n)$ is lin indep

pf of $\Rightarrow$: Suppose $\sum c_i v_i = 0$ i.e. $f\left(\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}\right) = 0$

But we know that $f\left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}\right) = 0$

$$\therefore \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \text{ as } f \text{ is 1-1}$$

pf of $\Leftarrow$: We need to show that f is 1-1

$$\text{Let } f\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right) = f\left(\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}\right) \text{ i.e.}$$

$$\therefore \sum a_i v_i = \sum b_i v_i \text{ i.e.}$$

$$\therefore \sum (a_i - b_i) v_i = 0$$

$$\therefore a_i = b_i \forall i \text{ as } v_i \text{ are lin indep}$$

B

Fixing an ordered basis for a f.d.m v.space V allows us to assign coordinates for any vector $v \in V$ with respect to that basis.

Suppose a finite dimensional v.space V has a basis $\{v_1, \dots, v_n\}$ and a basis $\{w_1, \dots, w_k\}$

We have isomorphisms $\mathbb{R}^k \cong V \cong \mathbb{R}^n$

$\therefore$ we have an isomorphism $\mathbb{R}^k \rightarrow \mathbb{R}^n$ (by composing)

As any basis of $\mathbb{R}^k$ has size k

$\left\{ \begin{array}{l} \text{---} // \text{---} \mathbb{R}^n \text{ ---} // \text{---} n \\ \text{basis goes to a basis under isomorphism} \end{array} \right\}$

we are forced to conclude $k = n$

Exercise

Left to justify: Why can't a finite dimensional V have an infinite basis. Do this using an isomorphism $V \cong \mathbb{R}^n$

More abstract proof of invariance of dimension of a f.d.v.s.p V

Claim (i) (Exchange property) Suppose $\begin{cases} u \in S \\ v \in \text{Span}(S) \text{ but} \\ v \notin \text{Span}(S - \{u\}) \end{cases}$

Then $u \in \text{Span}(S - \{u\} \cup \{v\})$

$\text{Span}(S - \{u\} \cup \{v\}) = \text{Span}(S)$

This is true but not directly useful as stated. We will use the same idea but in a slightly different way in the next class to finish the proof.

Reason: In any lin comb of elts of S that gives v , the coeff of u must be nonzero. So I can solve for u

Remark: If $A \subseteq B \subseteq V$, to

show that $\text{Span}(A) \subseteq \text{Span}(B)$, it is enough to show $A \subseteq \text{Span}(B)$

Consequence : Suppose $I = \{l_1, l_2, \dots, l_n\}$ is an indep set
and S spans V (for any v-space V)

Then $n \leq |S|$

Pf. We may assume S is finite (otherwise nothing to prove)

Let $S = \{v_1, \dots, v_k\}$

Want to show $n \leq k$. Suppose not. i.e. $n > k$

As $\text{span } S = V$, each $l_i \in \text{span}(S)$

idea { I will change S preserving its span & size
using exchange lemma. If $v_i \notin S$ then exchange
 v_i for something in S . Continue....

To be finished in the next class.